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Individual Assignment | Due 21st April 2026
Virginia Favero

Big Data and AI in Business
Prof. Max Darby

Shell plc

Individual Assignment

AI Project Plan: Governance Intelligence and Behavioural Nudging System at Shell

Trinity Business School
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Table of Contents

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Problem Definition

Following upon the data strategy proposed in Group 11's assignment, Shell plc's global operations face challenges in governance transparency, fragmented workforce data, and limited ability to influence employee behaviour toward sustainability goals, and Scope 3 emissions are particularly difficult to manage due to their indirect and distributed nature.

As outlined in the group assignment under the "HR Data" section, Shell faces significant governance and ESG challenges due to fragmented data across systems such as Workday, SAP, and project tools limits visibility into workforce decisions, reducing the organisation's ability to detect bias, ensure fairness, and maintain consistent ESG reporting. At the same time, employees lack real-time insights into the impact of their decisions, creating a behavioural gap where suboptimal choices persist.

To address this, and following the proposed implementation of a unified data strategy centred around a Global Person ID (GPID), Shell seeks to leverage artificial intelligence enabling both improved governance monitoring and behavioural influence toward fairer and lower-emission outcomes. Specifically, the CEO has mandated the development of an AI system that not only audits decision-making processes, but also actively influences them towards fairer and lower-emission outcomes.

This project therefore proposes an AI-driven Governance Intelligence and Behavioural Nudging System that integrates workforce data via GPID with ESG datasets. The system thus combines the components:

1. Governance Intelligence: bias detection, anomaly detection, data quality monitoring;
2. Behavioural Nudging: AI-driven recommendations for fairer and lower-emission decisions; and
3. Scalable Architecture: cloud-based data lake, feature store, API-based model serving.

This shifts the organisation from passive reporting to proactive decision optimisation, improving transparency, consistency, and sustainability outcomes.

Business Objective:

The primary **business objective** is to develop an AI system that detects bias and governance risks in workforce and ESG decision-making, while nudging employees toward fairer and lower-emission operational choices.

This objective addresses two key organisational challenges:

1. Governance inefficiencies due to fragmented data systems (Workday, SAP, Excel); and
2. Lack of behavioural incentives for employees to make sustainable decisions.

The system therefore combines monitoring (governance AI) with intervention (nudging AI), creating a feedback loop between insight and action.

To provide extra context, the proposed recommendation engine is grounded in behavioural science, particularly in the **Behavioural Nudge Theory** explored by Hoyt, Thaler, and Sunstein, which suggests that small changes in how choices are presented can significantly influence decision-making without restricting freedom. The key mechanisms applied are default effects, social norms, feedback loops, and framing effects, as seen more in detail in the table below:

Examples of Behavioural Nudges		
Nudge Type	**Examples**	**Why it Works**
Default Options	• Pre-select lower-emission suppliers • Default to diverse candidate shortlists • Suggest standardised salary bands	People tend to stick with default choices, reducing effort and bias
Real-Time Feedback	• Show emissions impact: e.g. "+12% vs alternatives" • Display fairness scores • Highlight budget vs ESG trade-offs	Immediate feedback influences decisions at the moment they are made
Social Norms	• Show peer behaviour e.g. "80% of managers chose lower-emission options" • Compare fairness metrics • Highlight top-performing teams	People align their behaviour with perceived norms and peer actions
Framing	• Present options as positive outcomes e.g. "reduces emissions by 15%" • Highlight fairness improvements	The way information is presented shapes decision-making
Reminders	• Prompt ESG checks before approvals • Periodic alerts on high-emission or biased patterns	Keeps sustainability and fairness considerations top-of-mind
Personalised Recommendations	• Suggest lower-emission suppliers or more diverse candidates based on past behaviour	Tailored suggestions are more relevant and effective
Commitment Nudges	• Encourage managers to set fairness or ESG targets	People are more likely to follow through on stated commitments

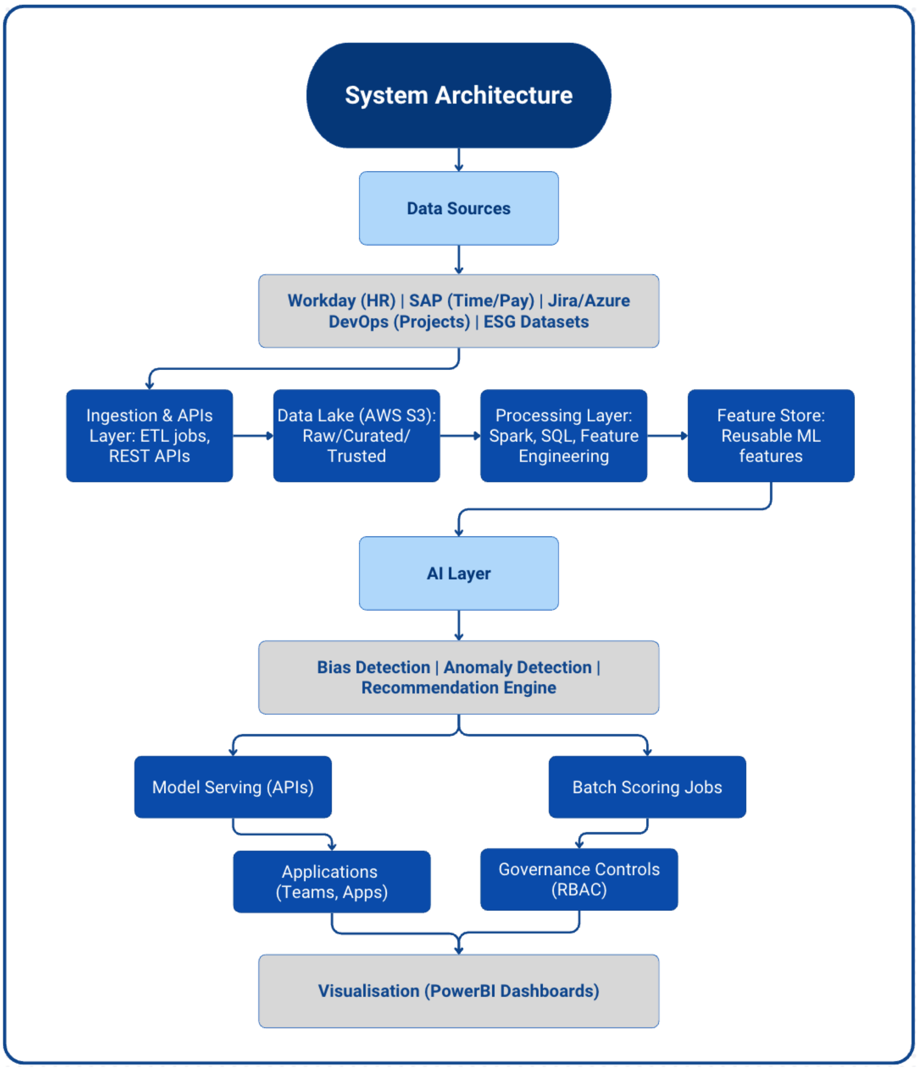
Data Strategy and Solution Overview

The solution is built on a unified data strategy centred on GPID, enabling integration of workforce and ESG data across systems such as Workday and SAP. Data is ingested via API-driven pipelines, standardised, and stored in a cloud-based data lake, with a feature store supporting reusable variables for AI models. The system is structured into four integrated layers, each supported by AI models suited to its function and data type, forming a layered hybrid architecture. To operationalise this strategy, the system is structured into four integrated layers:

AI Model Architecture by Layer					
Layer	Purpose	Key Functions	AI / Models Used	Rationale
Layer 1: Identity Layer	Unifies workforce data via Global Person ID (GPID)	Data integration, entity resolution, data quality checks	Record matching algorithms, data engineering pipelines, lightweight ML models (anomaly detection)	Handles structured relational data; focus is on data consistency rather than prediction, forming a reliable foundation for higher layers
Layer 2: Governance Intelligence Layer	Detects bias, governance risks, and ESG inconsistencies	Bias detection, anomaly detection, ESG monitoring, trend analysis	Feedforward (Dense) Neural Networks; anomaly detection models; time-series models (RNNs)	Structured HR and ESG data is best handled by dense networks; time-based models capture evolving governance patterns; CNNs are not suitable as data is not image-based
Layer 3: Behavioural Nudging Layer	Influences employee decisions in real time toward fairer and lower-emission outcomes	Recommendation generation, decision support, behavioural optimisation	Recommendation systems; Transformer-based models; Reinforcement learning	Combines contextual understanding (transformers) with adaptive learning (RL) to optimise interventions and improve outcomes over time
Layer 4: Visualisation Layer	Presents insights to stakeholders	Dashboards, alerts, reporting, performance tracking	No primary neural networks; uses outputs from upstream models	Focus is on interpretability and usability, ensuring AI outputs are actionable for managers, HR, and ESG teams

System Architecture

The system architecture can be extended using a multi-agent AI framework, where specialised agents perform distinct tasks such as bias detection, anomaly detection, recommendation generation, and user interaction. Each agent operates semi-independently but is arranged through a central planning component. For example: a Governance Agent that monitors bias and anomalies; a Nudging Agent that generates recommendations; or a User Interface Agent that communicates insights via natural language. This modular architecture improves scalability, explainability, and maintainability, while allowing the integration of different models optimised for specific tasks.



Stage 1: Model Development

The system incorporates Large Language Models (LLMs) to enhance behavioural nudging through context-aware recommendations. To improve reliability and reduce hallucinations, a Retrieval-Augmented Generation (RAG) approach is implemented, grounding outputs in verified internal ESG and HR data.

Business Metrics	
Governance KPIs	Including reduction in biased decisions, and improved auditability
Behaviour KPIs	Eg. adoption rate of AI recommendations
ESG KPIs	Encompassing reduction in emissions intensity per activity

Data Sources and Acquisition	
Workday	HR and employee data
SAP	Payroll, time-writing
Jira/Azure DevOps	Project contributions
ESG Datasets	Supplier emissions, lifecycle data

Data Preparation (Exploratory Data Analysis) and Feature Engineering:

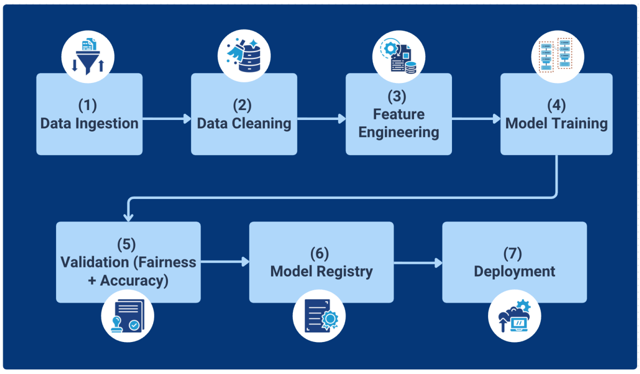
Three steps are followed to **prepare data** for the analysis, being: identity resolution via GPID, cleaning inconsistent records, and analysing bias patterns and emissions-intensive behaviours. The **variables** to analyse are then chosen, and examples include: promotion rates and pay progression under governance features, decision history, and supplier selection patterns as part of the behavioural features.

Training and Validation:

To implement the statistical and ML models, data should be divided into training and validation sets, where training data follows supervised models for bias and prediction, as well as unsupervised models for anomaly detection, while validation metrics include (but are not limited to) fairness metrics, precision and recall, and A/B testing for recommendations. Additionally, beyond the traditional metrics of precision and recall, model performance can be evaluated using task-specific benchmarks and behavioural outcomes, such as user adoption rates and decision improvements. For LLM components, evaluation approaches derived from benchmarks such as Massive Multitask Language Understanding (MMLU) can be adapted to assess the quality of generated recommendations.
For reference, MMLU is a leading, standardised evaluation for LLMs, measuring general knowledge and reasoning across 57 subjects (STEM, humanities, social sciences, etc.).

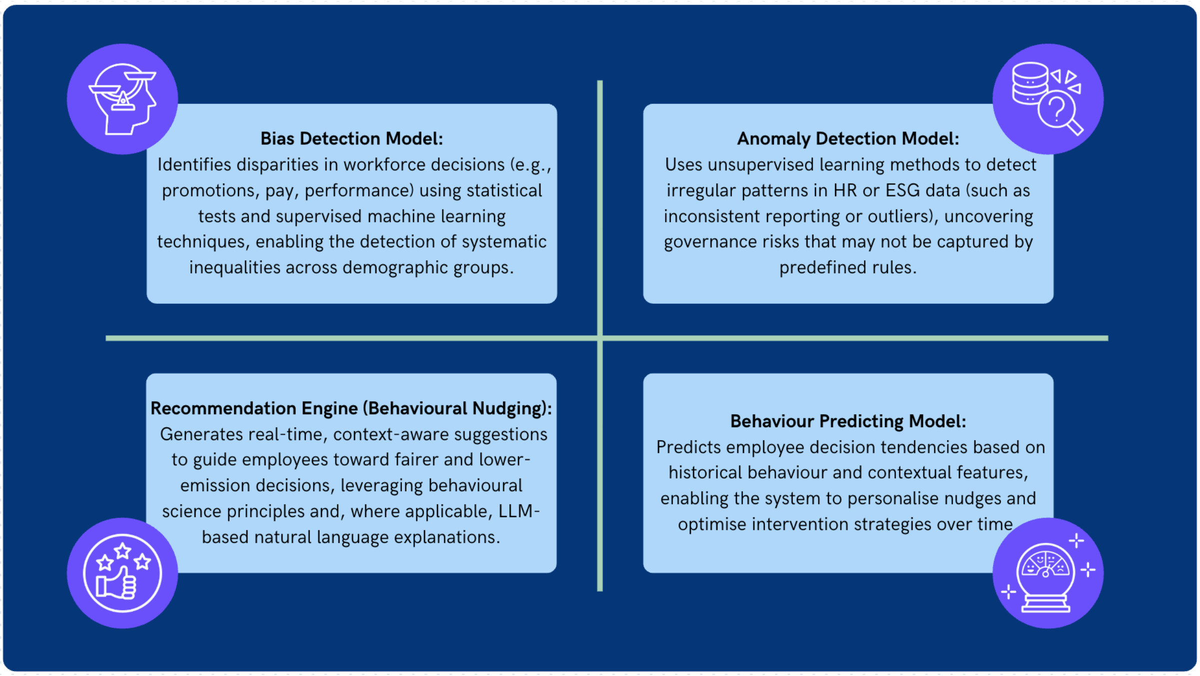
Development Pipeline:

The data is processed through an Extract, Transform, Load (ETL) pipeline via APIs, where it is ingested from multiple enterprise systems, cleaned, standardised, and integrated into a unified format. It is then feature engineered to capture key governance and behavioural signals, enabling the development and deployment of AI models for bias detection, anomaly identification, and real-time decision support.



Proof of Concept:

To validate feasibility, a **Proof of Concept (PoC)** should be conducted within a controlled business unit, such as a regional HR function or single business line. The PoC focuses on integrating a limited number of data sources, being for instance Workday and ESG datasets, through the GPID to demonstrate core system capabilities. Within the PoC, these models are deployed using a simplified architecture, like a subset of a cloud (like AWS) data lake and limited model pipeline, to minimise complexity while validating functionality. The PoC operationalises the system through four key AI models:



Success metrics of these models are measured through model performance, hence accuracy, precision, recall, and fairness, user engagement with nudges, and observable behavioural change, such as improved fairness scores or reduced emissions proxies.

Proof of Value:

Following the PoC, the **Proof of Value (PoV)** phase evaluates measurable business impact. This includes (estimated) governance improvements, through:

1. A 5-15% reduction in biased decisions as **governance impact**, with 20-40% faster audit/reporting processes.
2. **Enhanced operational efficiency**, with a 10-30% faster decision-making is due to real-time insights, and a 15-25% productivity improvement in HR workflows.
3. **Improved sustainability outcomes**, involving observable changes in employee decisions aligned with ESG targets, such as lower-emission project choices. There would be an improved alignment with Scope 3 targets with an estimated 3-10% reduction in emissions intensity per activity. This is because behavioural nudges influence supplier choice and operational decisions and small behavioural shifts can scale across thousands of employees.
4. **Data quality improvement**, translating to a 20-50% reduction in inconsistencies across HR systems, along with improved reporting accuracy and compliance. This is allowed by automated anomaly detection and unified GPID, together with continuous monitoring of data quality KPIs.

Additionally, A/B testing can be used to compare nudged versus non-nudged groups, quantifying the effectiveness of AI-driven recommendations, and ROI can be framed. The AI Governance and Nudging System creates value through cost reduction (efficiency and automation), risk reduction (bias, compliance fines), and ESG performance (long-term savings and reputation).

Budget Allignment:

The **budget allocation** aligns with strategic priorities in digital transformation, ESG compliance, and workforce optimisation. Here, a gradual implementation ensures more control over costs starting with high-impact areas before scaling globally. The table below illustrates examples of key cost components along with their justifications:

Budget Allignment			
Cost Component	**Description**	**Cost**	**Justification**
Cloud Infrastructure	AWS S3, compute, storage	• PoC: €10,000-€30,000 • Scaled (global): €200,000-€700,000/year	Large enterprises like Shell require: • Multi-region data storage • High compute for ML + LLMs • Continuous data ingestion (Workday, SAP, ESG feeds)
Data Engineering	ETL pipelines, API integrations	• PoC: €30,000 -€80,000 • Scaled: €150,000-€400,000/year	The system is data-heavy: • Integrating Workday + SAP + ESG is complex • Identity resolution (GPID) = non-trivial • Continuous pipelines required
Model Development & AI Components	Includes bias detection, anomaly detection, recommendation engine, LLM + RAG	• PoC: €50,000-€120,000 • Scaled: €300,000-€800,000/year	• Multiple models (not just one) • LLM integration (cost per usage + infra) • Continuous retraining + experimentation
Governance & Compliance	AI Ethics, EU AI Act, Bias Audits	• PoC: €15,000-€40,000 • Scaled: €120,000-€300,000/year	This is especially high for Shell because HR + ESG = high-risk AI (EU AI Act). Requires: • Bias audits • Documentation • Human oversight systems
Training & Change Management	Ensure employee adoption through AI literacy, communication, continuous user engagement	• PoC: €10,000 -€25,000 • Scaled: €80,000-€200,000/year	The system depends heavily on: • Employee adoption of nudges • Trust in AI decisions • Behavioural change (harder than tech)
Estimated Proof of Concept:	**€115,000-€295,000**	**Estimated Full Enterprise Rollout:**	**€850,000-€2.4M per year**

Sources for the approximations can be found in the References section.

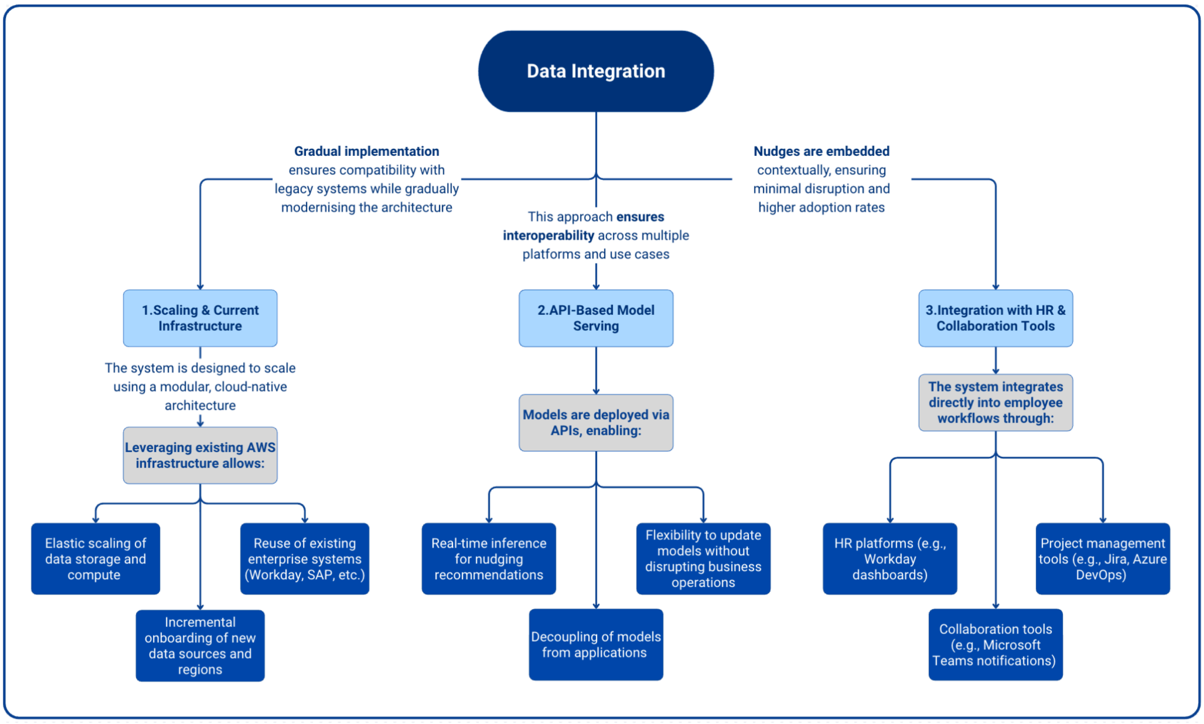
Stage 2: Deployment Pipeline

Ownership of models & Intellectual Property Rights (IPR):

Ownership of AI models and associated IPR should remain centrally governed to ensure consistency, compliance, and accountability. In particular, there should be an AI/data governance team responsible for models lifecycle management, where ownership can be divided by department, for example, HR owns workforce data and ESG teams own sustainability datasets. Further, regarding IPR, all models, features, and pipelines developed internally should remain proprietary assets, and in the case of the presence of third-party tools, clear contractual agreements defining data usage and ownership (especially with cloud providers) should be defined. Lastly, a governance board should oversee operations ranging from model approval and updates, ethical compliance, to risk management.

Integration:

The integration of data consists of three key steps and are expanded in the diagram below. The elements to achieve these steps are: (1) scaling and current infrastructure, where a gradual implementation ensures compatibility with legacy systems while gradually modernising the architecture; (2) API-based model serving, which ensures interoperability across multiple platforms and use cases; and (3) integration with HR and collaboration tools, where nudges are embedded contextually, ensuring minimal disruption and higher adoption rates.



The system can also integrate with Robotic Process Automation (RPA) tools to operationalise decisions. RPA bots can execute repetitive actions, such as updating HR systems or logging ESG decisions, while AI components provide intelligence and recommendations. This combination enables both automation and adaptability, where AI handles unstructured decision-making and RPA executes structured workflows.

People and Change Management:

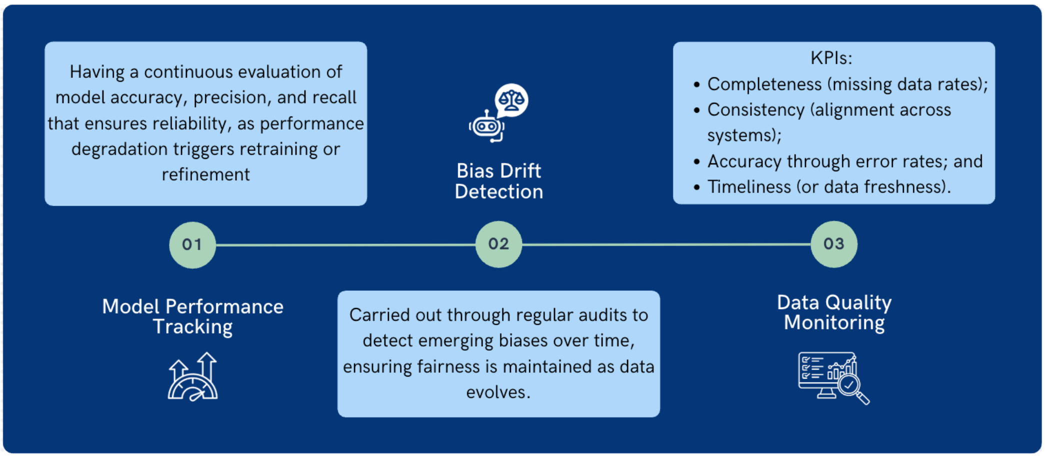
Considering the range of organisational stakeholders involved, three key steps should be followed. First, clear Service Level Agreements (SLAs) should be established to define system performance expectations, including model response time (real-time or near real-time), data pipeline refresh frequency, and system uptime and reliability aligned with accuracy thresholds. This step is crucial, as SLAs ensure accountability and alignment between technical teams and business stakeholders. Then, to ensure the adoption of the AI Governance and Nudging System, **educational and upskilling opportunities** should be incorporated. These can take the form of training programmes for HR, managers, and employees through AI literacy initiatives, tackling especially education on AI transparency to build trust and understanding, and through clear communication on how nudges work and their benefits. Thirdly, continuous **user testing** is essential in order to validate the usability of nudges and dashboards, identify potential future resistances or misunderstandings, and in general to improve the system design based on constructive feedback.

Stage 3: Monitoring and Maintenance

Current AI systems primarily operate at the level of "reckoning" rather than "judgement", meaning they can process and optimise decisions based on data but lack true understanding or accountability. Therefore, human oversight remains essential and consequently, a human-in-the-loop framework should be implemented, where critical decisions, particularly those affecting fairness, promotions, or ESG outcomes, require human validation. This ensures that the system remains aligned with organisational values and avoids over-reliance on purely statistical and probabilistic reasoning.

Monitoring:

There are three elements that can ensure effective monitoring: model performance tracking, bias drift detection, and data quality monitoring, as further described in the diagram below.



Maintenance:

Maintenance of the system also plays an important role. It would be accomplished mostly through a scheduling that is proportional to SLAs, and through continuous human-based feedback loops. In particular, maintenance cycles should be aligned with SLAs, including regular model retraining, data pipeline updates, and general system health checks. Instead, users' feedback loops should be integrated with UI, enabling refinement of nudges, identification of false positives/negatives, and ongoing system improvements.

Ethics and Governance

In order to properly implement the AI Governance and Nudging System, the themes of ethics and governance should be tackled and made tangible through different KPIs. Examples of KPIs for monitoring data quality are the levels of transparency, fairness, privacy, and accountability. Essentially, **transparency** means that the system incorporates explainable AI techniques, providing users with clear reasoning behind recommendations and decisions; **fairness** refers to having continuous bias audits to ensure that models do not reinforce existing inequalities, and fairness metrics are embedded into model evaluation. **Privacy** is ensured through GDPR-compliant practices such as role-based access control, data minimisation, and secure data handling. Lastly, **accountability** refers to having clear ownership structures to define responsibility for data (HR and ESG teams), models (AI governance team), and decisions (business stakeholders), as well as having audits that ensure traceability of all AI-driven decisions.

From a regulatory perspective, the system would likely fall under the "high-risk" category of the EU AI Act, as it impacts employment decisions and organisational governance. This requires strict compliance with transparency, accountability, and risk management obligations, including documentation, auditability, and human oversight. To operationalise transparency, explainable AI (XAI) techniques such as SHAP (SHapley Additive exPlanations) or LIME (Local Interpretable Model-agnostic Explanations) can be used to explain model outputs at both global and individual decision levels. However, it is important to recognise the limitations of XAI, particularly the distinction between explaining individual predictions versus overall model behaviour.

Evaluation and Discussion

AI Necessity vs Non-AI Alternatives:

While AI provides advanced capabilities, not all components strictly require it. For example, data integration and governance dashboards could be implemented using traditional rule-based systems and business intelligence tools. Similarly, emissions reporting could rely on deterministic carbon-accounting frameworks. Nevertheless, AI becomes necessary in detecting hidden bias across large and complex datasets, predicting behavioural patterns and future decisions, and generating personalised recommendations at scale. Hence, AI is not universally required but is justified where complexity, scale, and adaptability exceed human or rule-based capabilities.

Real-World Feasibility at Shell Scale:

Looking forward, similar AI governance systems could extend beyond digital workflows into physical AI and robotics systems, particularly in industrial environments. Advances in deep reinforcement learning and transformer-based architectures are enabling autonomous systems to make complex decisions in real-world environments. While not immediately applicable to Shell's HR and ESG use case, this highlights the broader trajectory towards integrated "decision-making systems" across both digital and physical operations.

The overall implementation of the AI Governance and Nudging System is **feasible but complex**, mainly due to global scale given the dimensions of a company like Shell and requires phased deployment. In fact, implementing this system at Shell presents significant challenges due to factors like global operations across multiple jurisdictions, legacy systems and inconsistent data quality, and possible cultural resistance to algorithmic oversight. Additionally, integrating ESG data with workforce systems requires cross-functional coordination that may not currently exist. Given that the reliance on GPID as a unified identity layer is critical yet its accomplishment is complex, having an incremental process to ensure feasibility is essential, starting with pilot projects in selected business units before scaling globally.

Tradeoffs - Accuracy vs Ethics:

Moreover, there are some **tradeoffs** that need to be accounted for, such as the balance between accuracy and privacy, or automation and trust. While on one hand a higher degree of accuracy leads to more detailed data, which subsequently improves model performance, and fine-grained tracking enables better predictions, ethical considerations have to be taken into account. In fact, increased data collection may raise privacy concerns, and detailed tracking risks resulting in employees' perceptions of surveillance. Having a balance ensures that technological capability does not compromise organisational trust or ethical standards.

Risks and Limitations:

Lastly, the main **limitations** of this system are the great dependency on data quality, potential user resistance, and "nudge fatigue". Particularly, bias detection models are probabilistic and may produce false positives, nudging systems may suffer from user disengagement (nudge fatigue), and data quality issues can significantly impact model reliability. Despite its challenges, the combined AI Governance and Nudging System represents a robust and forward-looking approach: its strength lies in integrating oversight with behavioural change, although careful implementation and governance are essential to realise its full potential.

Conclusion

This AI system enables Shell to transition from passive governance to proactive decision-making. By combining bias detection with behavioural nudging, the organisation can improve fairness, sustainability, and transparency simultaneously, and this approach ensures that the system does not merely analyse governance issues but actively improves decision-making behaviour across Shell. While challenges exist, it represents a scalable and forward-looking approach to aligning workforce decisions with ESG goals, as this system ultimately supports Shell's broader ESG ambitions while strengthening internal trust and accountability.

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